

Venue Daily Visitors

Data Analysis Report

Data Visualisation Module | Student ID: 001496957

1. Introduction to Data Visualisation

Data visualisation is the practice of representing numbers and records as charts, graphs and maps so that patterns, trends and outliers become visible to the human eye. A large spreadsheet of daily visitor counts across 35 venues is almost impossible to interpret by reading the raw figures. The right chart can reveal a seasonal pattern, an unusual venue, or a relationship between two variables in seconds, where reading the data directly might take hours.

Several types of chart are used in this report. A **bar chart** compares totals across a set of categories. Sorting the bars in descending order and applying consistent colours makes the most important categories stand out immediately. A **line chart** shows how a value changes over time. When daily data is noisy, applying a **rolling average** smooths out short-term variation so that the underlying trend becomes clear. Adding a **trendline** using linear regression shows whether a venue is growing, declining or staying flat across the year. A **box plot** summarises the full distribution of a variable in a compact form, showing the median, the middle 50% of values, and any outlier days. A **heatmap** uses colour intensity to show a third value across two categorical axes at once. A **scatter plot** places two numerical variables on the two axes so that any relationship between them becomes visible as a pattern of points.

Two of the eight charts are **interactive**, built using the hvplot and panel libraries. They allow the reader to zoom, pan, hover over individual data points to see exact values, change the smoothing window, switch between venue groups and reassign the axes. The eight visualisations are ordered to tell a logical story, starting with a high-level overview of all 35 venues, then narrowing to time trends, daily distributions, attribute relationships and finally the interactive exploration tools.

2. Findings

Figure 1 - Total Annual Visitors by Venue

Justification. The natural first step is a simple overview of how much each venue contributed across the year. A bar chart is the clearest way to compare a single quantity across many categories, and sorting the bars in descending order means the eye goes immediately to the highest performers. Colour shows which volume tier each venue belongs to, so the reader sees both the individual total and the group it belongs to without reading every label. This follows the approach in the lecture file **L02 / 08BarChart all sorted.py**, which demonstrated sorting before plotting and applying category-based colour coding. The adaptation involved replacing page hit data with venue visitor totals and using a four-colour tier scheme.

What Figure 1 shows. Visitor numbers are very unevenly distributed. A small number of green High-volume venues attract far more visitors than all the others combined. Most venues fall into the red Low category, forming a long tail of modest performers. Several dark grey Very Low venues registered barely any visitors across the whole year, raising a question about whether they were operating for the full year. That question is followed up in Figure 2.

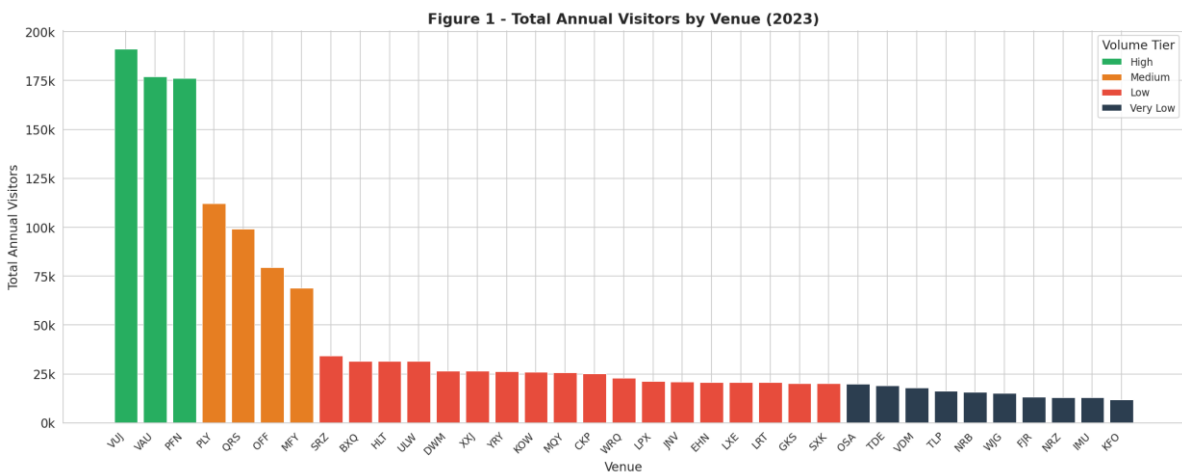


Figure 1 - Total Annual Visitors by Venue (2023)

Bars sorted in descending order. Colour indicates volume tier: green = High (>150,000), orange = Medium (>48,000), red = Low (>20,000), dark = Very Low. Source: L02 / 08BarChart all sorted.py

Figure 2 - Annual Visitors by Volume Tier

Justification. Figure 1 shows all venues together but the scale difference makes intra-tier comparison difficult. Plotting each tier in its own panel gives each group its own scale so differences within each group become visible. The venue lists were generated automatically using a threshold-based loop, following **L02 / 06BarChart automatic.py**, which demonstrated classifying pages into High, Medium and Low groups using a for loop with numeric thresholds rather than manually specifying which venues go in which group. I adapted this into a 2x2 subplot grid so all four panels are visible at once.

What Figure 2 shows. Within each tier the venues are not equal. The High panel shows one or two venues clearly outperforming the others. The Medium panel has a wider spread. The Very Low panel is particularly interesting: some venues have near-zero totals, which is more likely to mean they opened partway through 2023 than that they simply performed poorly. This matters because it means their annual totals should not be compared directly to venues that were open for the full year.

Figure 2 - Annual Visitors by Volume Tier

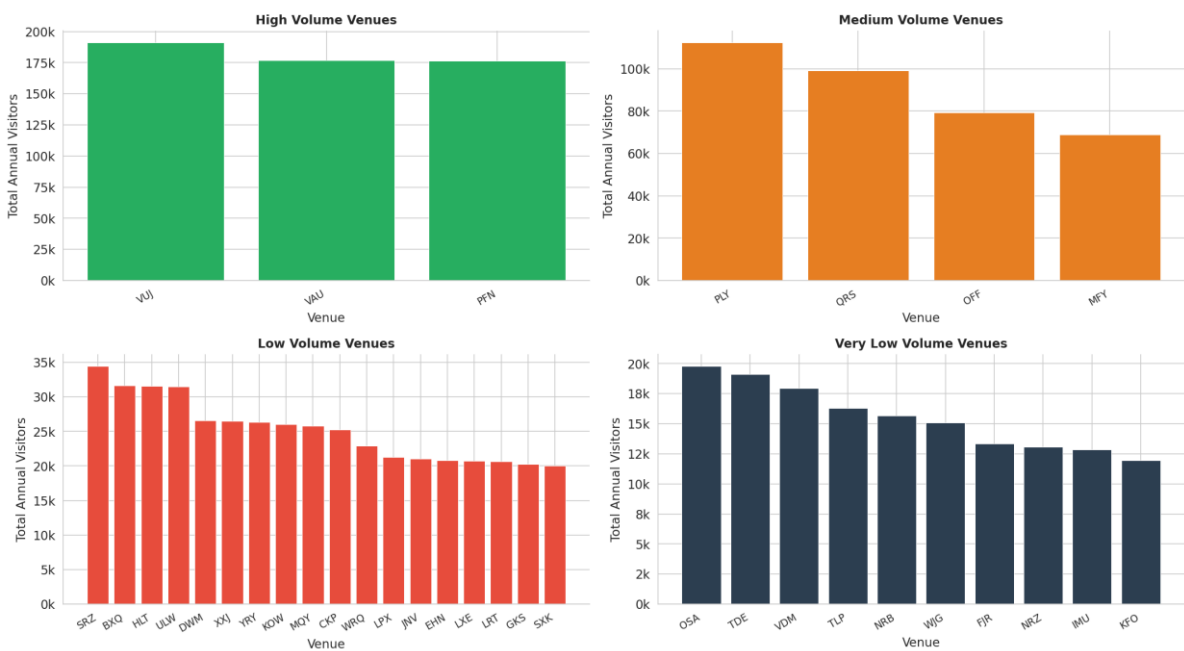


Figure 2 - Annual Visitors by Volume Tier

Each panel shows one tier. Bars sorted within each panel. Source: L02 / 06BarChart automatic.py (threshold-based automatic classification).

Figure 3 - Rolling Average and Trendline

Justification. Having identified which venues matter most in Figures 1 and 2, the next question is how their visitor numbers changed over 2023. Raw daily data is too noisy to read clearly. The solution from **L03 / 05LinePlot high volume rolling ave.py** is to apply a rolling average using `data.rolling(window=21).mean()`, which replaces each value with the mean of the surrounding 21 days. I then added a linear trendline using `np.polyfit(x, y, 1)` and `np.poly1d(z)` as shown in **L03 / 06LinePlot high volume trendline.py**. The thin faded lines in the background show the raw daily data so the degree of smoothing is transparent.

What Figure 3 shows. High-volume venues have nearly flat trendlines, meaning their popularity stayed stable throughout 2023. There is a small shared dip in the rolling averages mid-year, likely a seasonal effect such as school holidays. Medium venues are more varied: some have upward trendline slopes indicating genuine growth, others are flat or slightly declining. This difference in behaviour was not visible in Figures 1 or 2.

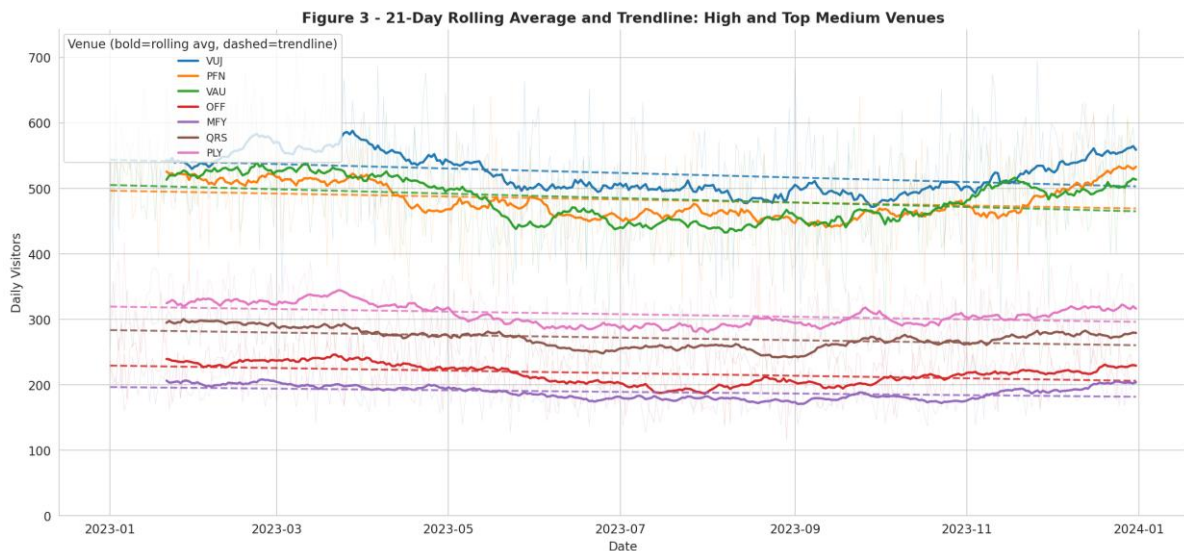


Figure 3 - 21-Day Rolling Average and Linear Trendline: High and Top Medium Venues

*Bold lines show the 21-day rolling average. Thin faded lines show raw daily data. Dashed lines show the linear trendline.
Source: L03 / 05LinePlot rolling ave.py + 06LinePlot trendline.py.*

Figure 4 - Box Plots: Daily Visitor Distribution

Justification. The rolling average shows trends over time but not how consistent each venue is day to day. A box plot from `L05 / 04BoxPlot high volume.py` summarises the full distribution of daily visitor counts in a compact form. The box spans the middle 50% of values, the line inside shows the median, the whiskers extend to 1.5 times the interquartile range, and outlier days appear as dots. I adapted the lecture code to show High and Medium venues side by side so the contrast between the two tiers is directly visible.

What Figure 4 shows. High-volume venues generally have taller boxes, meaning their daily visitor counts vary more. This is most likely driven by strong weekend versus weekday differences at popular venues. Medium venues tend to have narrower boxes, indicating more consistent day-to-day numbers. The outlier dots above the whiskers on several venues represent exceptional days, probably bank holidays, events or promotional periods.

Figure 4 - Daily Visitor Distribution: Box Plots by Volume Tier

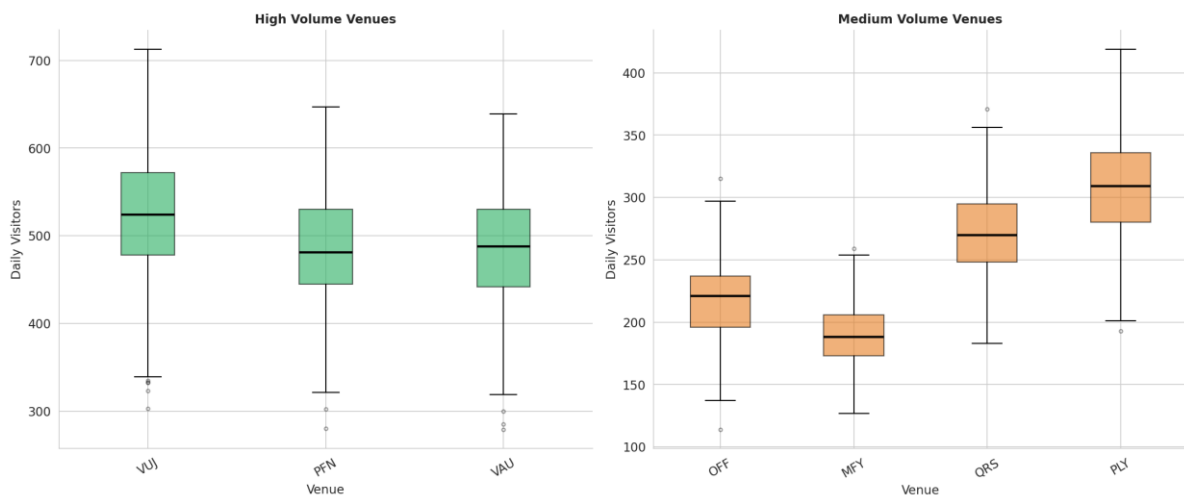


Figure 4 - Daily Visitor Distribution: Box Plots for High and Medium Volume Venues

The box shows the middle 50% of daily values. The line inside is the median. Dots above the whiskers show outlier days.
Source: `L05 / 04BoxPlot high volume.py`.

Figure 5 - Correlation Heatmap

Justification. The summary dataset has six numerical attributes per venue. Checking all possible pairings with individual scatter plots would require 15 separate charts. A correlation heatmap from **L04 / 04HeatMap correlation all.py** calculates all pairwise Pearson correlations at once using `data.corr()` and displays them as a colour grid. Red means the two attributes tend to increase together. Blue means they move in opposite directions. White means no linear relationship. The numbers in each cell give the exact coefficient. I adapted the code by applying it to the six summary attributes.

What Figure 5 shows. The strongest positive correlation is between Spend and Duration. Venues where visitors stay longer tend to see higher average spend per visit. There is also a moderate negative correlation between Age and TotalVisitors, meaning venues with an older average visitor age tend to attract fewer visitors overall. Distance and FemalePercent show only weak correlations with the other variables. The Spend-Duration relationship is examined more closely in Figure 6.

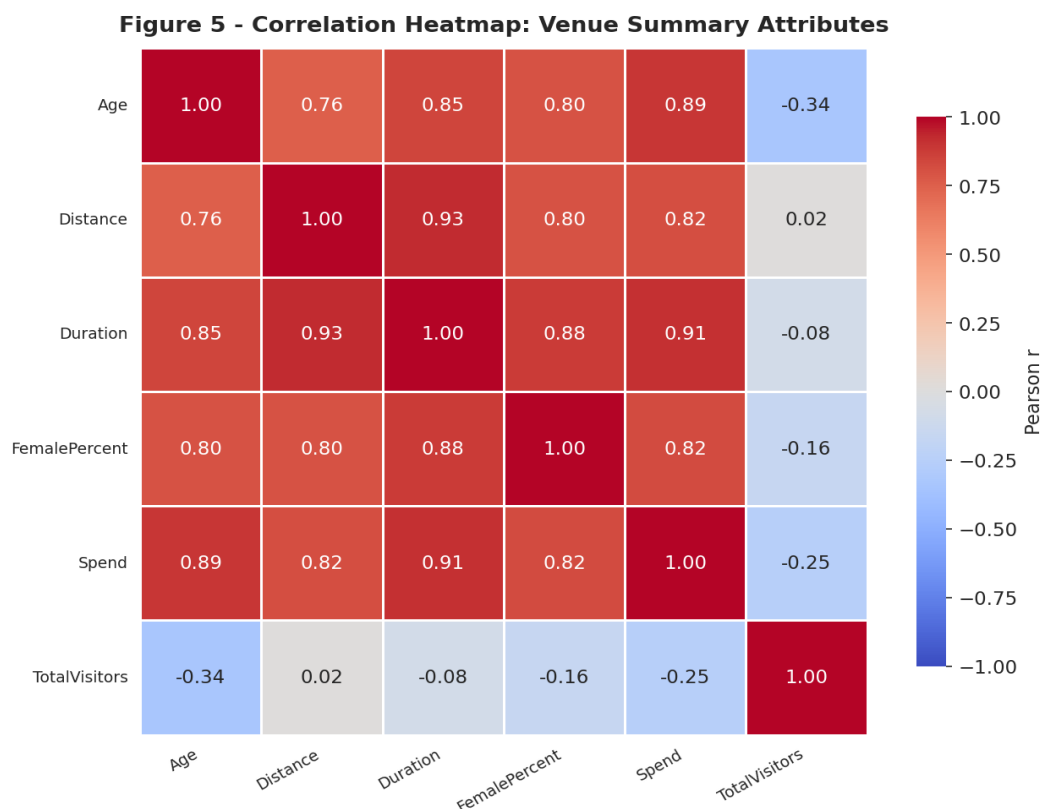


Figure 5 - Correlation Heatmap: Venue Summary Attributes

Red = positive correlation, blue = negative, white = no relationship. Numbers show the Pearson r value. Source: L04 / 04HeatMap correlation all.py.

Justification. The heatmap in Figure 5 identified a positive correlation between Spend and Duration but a single number does not show the shape of that relationship or which venues are driving it. A scatter plot from **L04 / 01ScatterPlot high volume.py** places each venue as an individual labelled point, with Duration on the x-axis and Spend on the y-axis. Each point is labelled with the venue code, following **L04 / 07ScatterPlot selected correlation.py**. Points are colour-coded by tier. A linear trendline computed with `np.polyfit` is added across all venues to confirm the direction of the relationship.

Figure 6 - Average Spend vs Visit Duration per Venue

This scatter plot illustrates the relationship between the average spend per venue and the average visit duration. The data is segmented into four volume tiers, showing how spending behavior varies across different levels of customer frequency.

Legend:

- High (Green)
- Medium (Orange)
- Low (Red)
- Very Low (Dark Blue)
- trendline (Dashed grey line)

Data Points (Approximate Values):

Venue	Volume Tier	Visit Duration (min)	Average Spend (£)
LXE	Low	65	10
EHN	High	68	9
PFN	High	70	9
PLY	Medium	72	11
MU	Very Low	75	12
YRY	Low	78	15
CKP	Low	80	13
MQY	Low	82	14
WJG	Very Low	85	12
OSA	Very Low	88	19
SRZ	Low	90	17
KOW	Low	92	16
WRQ	Low	95	21
GKS	Low	98	20
XXJ	Low	95	32
ORS	Medium	100	22
VAU	High	105	18
SXK	Low	110	27
LPX	Low	112	26
BXQ	Low	115	28
HLL	Very Low	118	30
TLP	Very Low	120	31
OFF	Medium	118	14
VDM	Very Low	125	35
DWM	Low	130	36
MFY	Medium	132	38
NRZ	Very Low	135	33
LRT	Low	140	42
INV	Low	142	37
NRB	Very Low	145	39
FIR	Very Low	148	44
VUJ	High	142	29
TAL	Very Low	150	40
KFO	Very Low	152	41

Each point is one venue, labelled with its code and coloured by tier. Dashed line is the overall linear trendline. Source: L04 / 01ScatterPlot + 07ScatterPlot selected correlation.py.

Figure 7 - Interactive Rolling Average Explorer

Justification. Figure 3 was limited to a fixed 21-day window and fixed venue selection. The interactive version built using `hvplot.line()` and panel widgets, following [L07 / 02LinePlot high volume.py](#), gives the reader three controls: a dropdown to switch between High, Medium and Low tiers; an integer slider to change the rolling window from 1 to 60 days; and a date range slider to zoom into any period. The Bokeh toolbar also provides pan, box zoom, hover, save and reset tools. This makes it possible to explore patterns at different time scales without rewriting any code.

What Figure 7 shows. At a 7-day window a weekly rhythm becomes visible: certain days of the week consistently attract more visitors, almost certainly a weekend effect. At 21 days that pattern smooths out and the monthly trend becomes clearer. At 60 days only the broad seasonal shape remains. Switching from High to Medium venues reveals far more divergence between venues than Figure 3 showed, with some Medium venues showing clear growth and others declining. The hover tool shows the exact visitor count for any date.

How to use Figure 7 (in the notebook): Use the Volume Tier dropdown to switch venue groups. Drag the Rolling Window slider left for less smoothing or right for more. Use the Date Range slider to narrow to a specific period. Use the toolbar to pan, zoom, hover or save the chart.

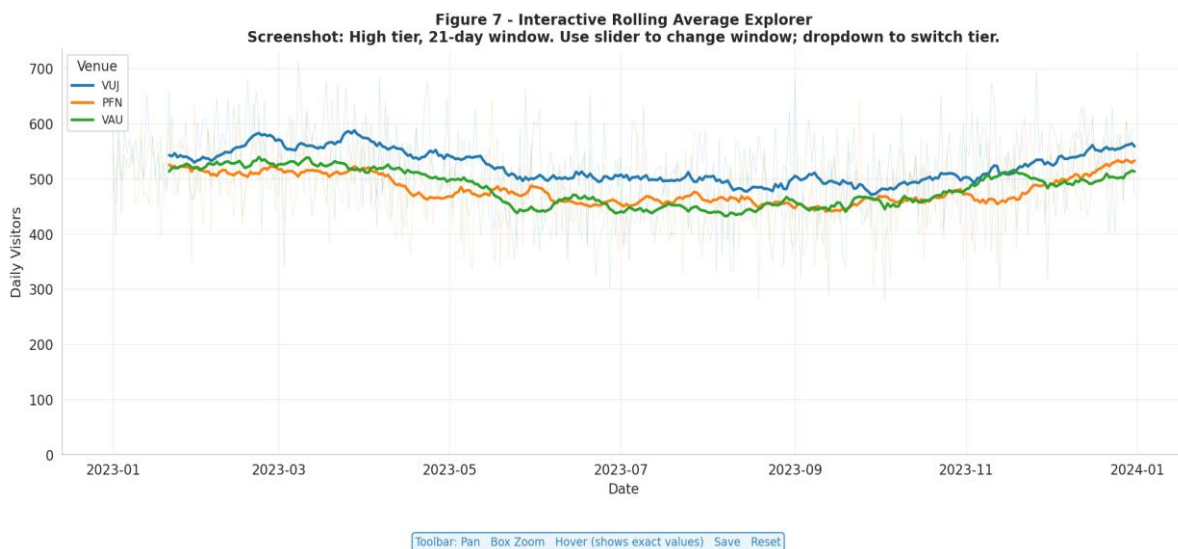


Figure 7 - Interactive Rolling Average Explorer (screenshot: High tier, 21-day window)

Screenshot from the Jupyter notebook. Slider changes the rolling window (1-60 days). Dropdown switches between tiers. Toolbar: Pan, Box Zoom, Hover, Save, Reset. Source: [L07 / 02LinePlot high volume.py](#).

Figure 8 - Interactive Scatter Explorer

Justification. Figure 6 showed only one attribute pairing. With six numerical columns there are 15 possible pairings. Rather than producing 15 static scatter plots, an interactive version was built using `hvplot.scatter()` and panel Select widgets, following **L07 / 09ScatterPlots selected correlation.py**. Two dropdowns let the user choose which attribute goes on each axis, updating the chart instantly. The hover tool shows the venue code and exact values for any point. Points are colour-coded by tier throughout, keeping the colour scheme consistent with every other figure.

What Figure 8 shows. Switching the x-axis to Age and keeping TotalVisitors on the y-axis confirms the negative relationship identified in Figure 5: venues with an older average visitor age attract fewer visitors overall. Changing to Distance vs TotalVisitors shows no clear pattern, confirming that travel distance does not predict how busy a venue will be. The hover tool makes it easy to identify which specific venue any outlier point represents, so it can be cross-referenced with Figures 1 and 3.

How to use Figure 8 (in the notebook): Use the X Axis and Y Axis dropdowns to change what is plotted on each axis. Hover over any point to see the venue code and exact attribute values. Use Box Zoom to zoom into a cluster. Click Reset to return to the full view.

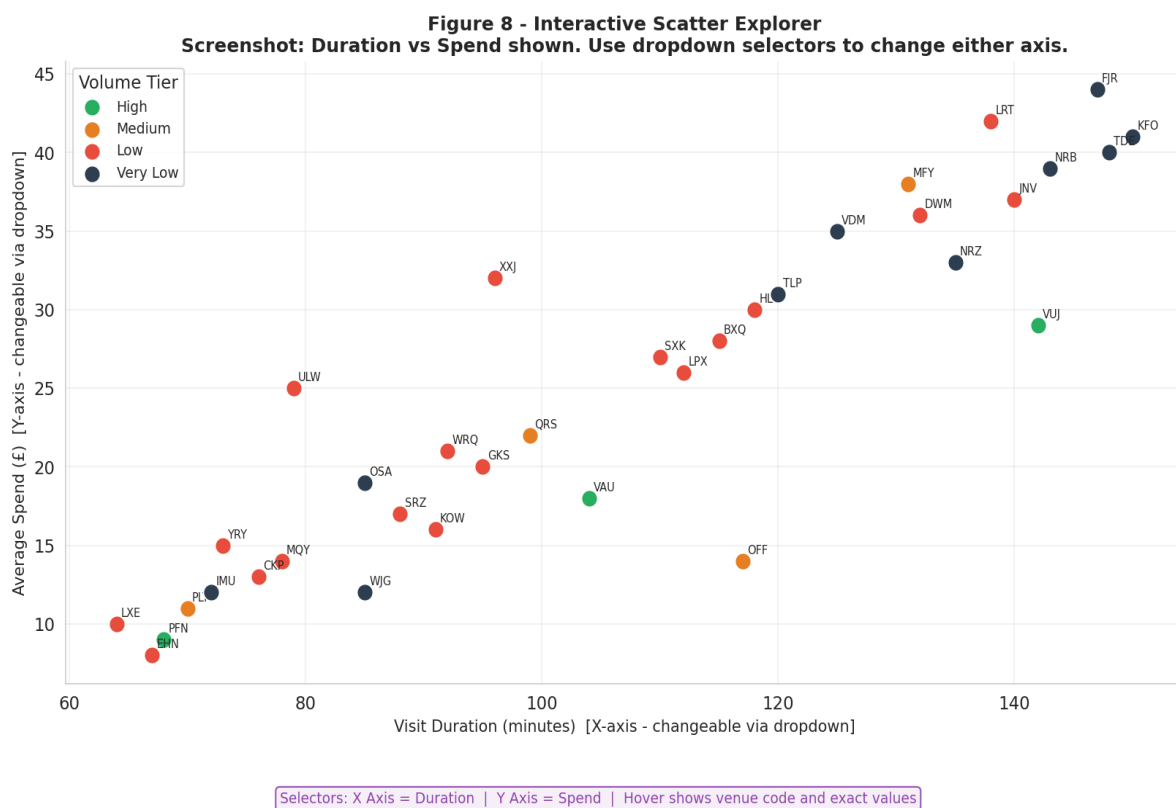


Figure 8 - Interactive Scatter Explorer (screenshot: Duration vs Spend)

Screenshot from the Jupyter notebook. Dropdown selectors reassign both axes to any of the six summary attributes. Hover shows venue code and values. Source: `L07 / 09ScatterPlots selected correlation.py`.

3. Critical Review

Working through this dataset gave me a clearer sense of why the sequence of techniques in the module follows the order it does. Starting with proportion before moving to time, then correlation, distribution and finally interaction reflects the natural order in which questions arise during exploration. You cannot meaningfully analyse trends for specific venues until you have first identified which venues are worth focusing on. You cannot interpret a correlation until you have understood what each variable looks like on its own.

One of the most useful things I applied was keeping a consistent colour scheme across all eight figures. Green always means High, orange always means Medium, red always means Low and dark means Very Low. This meant I only had to establish that colour coding in Figure 1 and the reader could then interpret every subsequent chart without re-reading the legend. This is a best practice mentioned in the lectures and its value was clear once the notebook was complete.

The rolling average technique from Lecture 03 was probably the most impactful single technique in this analysis. The raw daily data was nearly unreadable as a line chart. With a 21-day window the underlying trends became clear immediately. Combining that with the trendline showed not just where the trend was but whether it was moving up, down or staying flat. The box plots from Lecture 05 added something the time series charts could not show: how variable each venue was within a period, not just what the average looked like over time.

There are some limitations worth acknowledging. This analysis covers only 2023, so it is impossible to tell whether the seasonal dip in Figure 3 is a recurring annual pattern or something specific to that year. The venue codes are anonymous, which makes it hard to explain some of the relationships. For example, the Spend-Duration correlation could simply reflect that venues designed for longer visits happen to have more food and retail options, rather than dwell time causing higher spend in a way that could be deliberately engineered. The summary attributes are also means, which hides the distribution within each venue.

4. Conclusions

The main findings from the eight visualisations are:

- Visitor numbers are very unevenly distributed across the 35 venues. A small number of High venues account for a disproportionate share of total footfall (Figures 1 and 2).
- High-volume venues maintained stable visitor numbers throughout 2023. Their rolling average trendlines are nearly flat. Medium venues showed more variation, with some growing and some declining (Figure 3).
- Some Very Low venues had near-zero visitors at the start of 2023, suggesting they opened partway through the year. Their annual totals should not be compared directly to venues open for the full year (Figure 2).
- High venues have wider day-to-day variation in visitor numbers than Medium venues, likely driven by strong weekend versus weekday differences. Outlier dots in the box plots indicate exceptional event days (Figure 4).
- Spend and visit duration are positively correlated. Venues where visitors stay longer also see higher average spend per visit. This holds across all four volume tiers (Figures 5 and 6).
- Venues with an older average visitor profile tend to attract fewer total visitors. This was identified in the correlation heatmap and confirmed interactively in Figure 8.
- Travel distance and gender split show weak correlations with all other attributes and appear to have little predictive power for either visitor volume or spend in this dataset (Figure 5).